AHUL261: Introduction to Psychology

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Why should engineers study psychology?





COVID PANDEMIC

the biggest health emergency of our generation

Vaccines and Behavior change

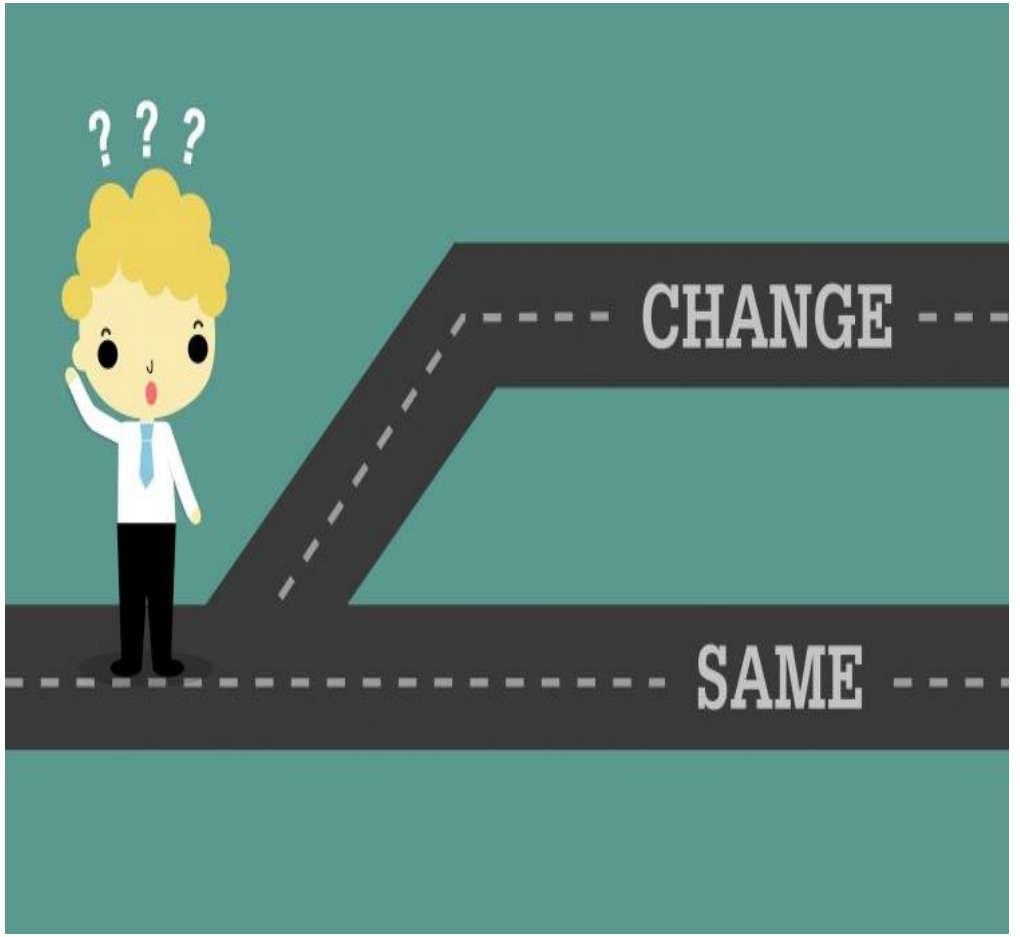
- Even with the vaccines, an important way to restrict the spread of the virus is restriction of the movement of people.

- There are many issues in vaccine delivery – misinformation, availability, distribution, hesitancy.

- Therefore, we must depend on behavioral changes and hence, upon human psychology for controlling the spread of infection.

Behavior change was critical in the Pandemic

- Lockdown: people must stay at home in order to flatten the curve of infection, the effectiveness of the policy depends on whether or not they do.
- A fully compliant population could reduce the proportion of those infected.
- Thus, many government-led campaigns to encourage people to '*Stay at Home and Save Lives*'. Behavior, then, is clearly critical.
- Indeed, controlling the virus also requires getting people to behave appropriately, wear masks — to 'do the right thing'.



Why Psychology Matters

- The trajectory of COVID-19 pandemic was critically dependent on psychological considerations.
- Without psychology we could have suffered far greater loss of human life.
- **Psychology saved lives in covid pandemic.**
- So many aspects of our individual and collective life, similarly, require understanding of human thinking and behavior.
- Psychology can be immensely helpful in your personal and professional journey

What is Psychology?

What is Psychology?

- Psychology - Greek Origin of the word

  Psyche: "breath, spirit, soul, mind"
  Logos: Knowledge or study

- Was part of philosophy until 19th century.

- Earlier definitions of Psychology – science of soul….. Science of mind

- Contemporary Definition: **The scientific study of behavior and mental processes**

- Like physics studies matter and biology studies living organisms, psychology studies behaviour and mental processes using scientific methods.

Three important aspects of Psychology

1. Psychology seeks to understand **behaviour** –

Behavior: Overt (i.e., can be directly observed, as with crying)

2. Psychology studies **mental processes** –

mental Processes: Covert (i.e., cannot be directly observed, as with remembering)

3. Psychology is a **Science**

Psychology as a Science

Defining Science

- Science is a body of systematised knowledge.

- Science came into existence as a way of answering questions about nature by examining nature directly rather than by depending on religious dogma, past authorities, superstition, or abstract thought processes alone.

- Two major components of science –

(1) Empirical observation

(2) Theory

Observation and Theory

- Empirical observation - Information gathered by systematically observing, measuring, or collecting evidence from the real world.

- Theory - A coherent set of ideas that explains why and how observations occur and makes predictions that can be tested.

- Observation tells us *what* happens. Theory tells us *why* it happens.

Example

- **Empirical observation -** Students who receive encouraging feedback from teachers tend to participate more in class.

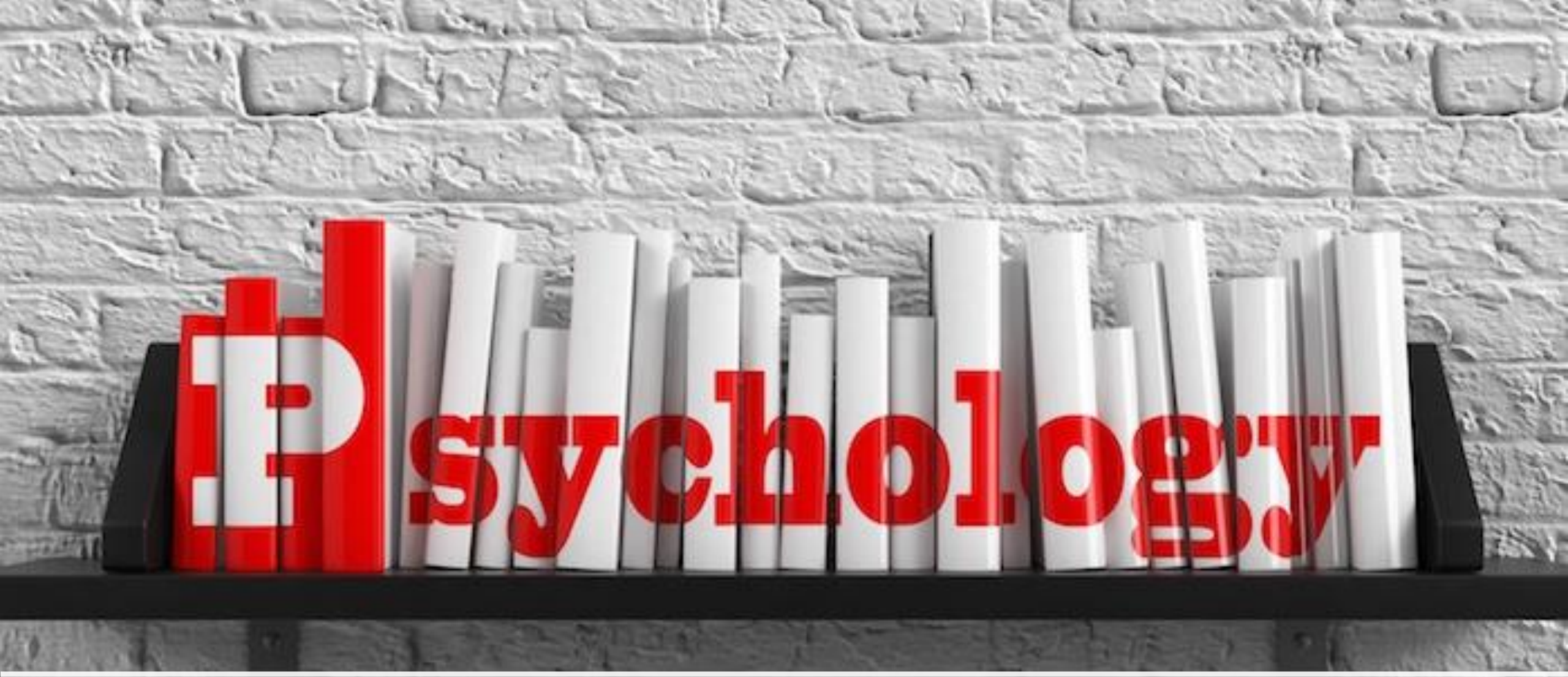
- **Theory** - Self-efficacy theory in Psychology proposes that believing in one's ability increases motivation and persistence. Positive feedback strengthens this belief.

- **Analogy:**
  - **Empirical observations** are like **sensor readings** from a machine.
  - **Theory** is like the **engineering model** that explains how the machine works and predicts its behavior under different conditions.

Scientific Method in Psychology

Observation → Pattern → Theory → Prediction → New Observation

- Empirical observations are the bricks; theories are the blueprint.
- Psychology advances by continually using evidence to build, test, and improve its explanations.



The Subfields of Psychology: Psychology's Family Tree

Subfields of Psychology

- Understanding sources of change and stability in behavior
  - **Developmental psychology**
    - Changes in behavior over the lifespan (womb to tomb)
  - **Personality psychology**
    - Examines consistencies in people's behavior over time and traits that differentiate us from one other

Subfields of Psychology

- Physical and mental health
  - **Health Psychology**
    - Explores relationship between psychological factors and physical ailments or disease
  - **Clinical Psychology**
    - Investigates diagnosis and treatment of psychological disorders
  - **Counseling Psychology**
    - Focuses on educational, social, and career adjustment problems

Subfields of Psychology

- Understanding our social networks
  - **Social psychology**
    - Studies how people's thoughts, feelings and behavior are affected by others
  - **Cross-cultural psychology**
    - Focuses on the similarities and differences in psychological functioning across cultures and ethnic groups

Subfields of Psychology

- New frontiers
  - **Engineering psychology**
    - the science of human behavior and capability applied to the design and operation of systems and technology.
  - **Behavioral Genetics**
    - How we might inherit certain behavioral traits and how the environment influences whether we actually display such traits.
  - **Clinical neuropsychology**
    - Focuses on relationships between biological factors and psychological disorders

The Education of a Psychologist

Doctoral Degrees

- **Ph.D (Doctor of Philosophy)**
  Research-focused; prepares for careers in research, academia, and advanced scholarship.
- **Psy.D (Doctor of Psychology)**
  Practice-focused; emphasizes clinical training and preparation for professional practice.

Master's Degree

Builds specialized knowledge and skills in areas such as counseling, clinical psychology, I/O psychology, or research methods.

Bachelor's Degree

Provides a foundation in psychology—introducing key concepts, research methods, statistics, and applications of psychology.

What differentiates psychologists and psychiatrists?

Psychologists

- Advanced training in psychological assessment, therapy, research, and behavior.
- **Do not** prescribe medication.
- Focus on understanding and treating mental, emotional, and behavioral issues.

Psychiatrists

- Medical doctors (MD/MBBS) with specialized training in mental health.
- Can diagnose, prescribe medication, and manage medical aspects of mental illness.
- Often use a combination of medication and other treatments.

By the end of this course, you should be able to answer questions like:

- Why do we think, feel, and behave the way we do?
- How do our minds, our social environments, and our biology shape our behaviour?
- What can psychological science tell us about health, learning, decision-making, leadership, use of technology, prejudice, and wellbeing?

End